

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 05 · Arithmetic Sequences

Questions

Student book · questions with writing space

- 17 pages
- Past papers up to January 2026

PURE 2 | TOPIC 05

Arithmetic Sequences

Your classified route

6

QUESTIONS

41

MARKS

2

FAMILIES

3

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 05 | FAMILY 01

Unknown Terms

WHAT THIS FAMILY ASKS

Recognise the unknown terms demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 Use simultaneous equations when two term positions are given.

2 classified questions

14 total marks

8. In a large theatre there are n rows of seats, where n is a constant.

The number of seats in the first row is a , where a is a constant.

In each subsequent row there are 4 more seats than in the previous row so that

- in the 2nd row there are $(a + 4)$ seats
- in the 3rd row there are $(a + 8)$ seats
- the number of seats in each row form an **arithmetic** sequence

Given that the **total** number of seats in the first 10 rows is 360

(a) find the value of a .

(2)

Adaptive unit 1 · WMA12-01_2023_Oct_Q08

DR ESAM AHMED

8. In a large theatre there are n rows of seats, where n is a constant.

The number of seats in the first row is a , where a is a constant.

In each subsequent row there are 4 more seats than in the previous row so that

- in the 2nd row there are $(a + 4)$ seats
- in the 3rd row there are $(a + 8)$ seats
- the number of seats in each row form an **arithmetic** sequence

Given that the **total** number of seats in the first 10 rows is 360

Given also that the total number of seats in the n rows is 2146

(b) show that

$$n^2 + 8n - 1073 = 0$$

(2)

Adaptive unit 2 · WMA12-01_2023_Oct_Q08

DR ESAM AHMED

8. In a large theatre there are n rows of seats, where n is a constant.
The number of seats in the first row is a , where a is a constant.
In each subsequent row there are 4 more seats than in the previous row so that

- in the 2nd row there are $(a + 4)$ seats
- in the 3rd row there are $(a + 8)$ seats
- the number of seats in each row form an **arithmetic** sequence

Given that the **total** number of seats in the first 10 rows is 360

(c) Hence

- (i) state the number of rows of seats in the theatre,
- (ii) find the maximum number of seats in any one row.

(3)

Adaptive unit 3 · WMA12-01_2023_Oct_Q08

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2. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

In an arithmetic series,

- the sixth term is 2
- the sum of the first ten terms is -80

For this series,

(a) find the value of the first term and the value of the common difference.

(4)

Adaptive unit 1 · WMA12-01_2024_MayJune_Q02

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2. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

In an arithmetic series,

- the sixth term is 2
- the sum of the first ten terms is -80

For this series,

(b) Hence find the smallest value of n for which

$$S_n > 8000$$

(3)

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TOPIC 05 | FAMILY 02

Series Sum

WHAT THIS FAMILY ASKS

Recognise the series sum demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 If last term is unknown, use $l = a + (n-1)d$ first.

4 classified questions

27 total marks

8. (i) (a) In an **arithmetic** series the first term is a and the common difference is d .

Show that

$$S_n = \frac{n}{2}\{2a + (n - 1)d\} \tag{3}$$

(b) Hence find

$$900 + 892 + 884 + \dots + 500 \tag{3}$$

Adaptive unit 1 · WMA12-01_2024_MayJuneR_Q08

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(4)

1. The arithmetic series S is given by

$$S = 2 + 5 + 8 + 11 + + 254$$

Find

(a) the number of terms in the series,

(2)

Adaptive unit 1 · WMA12-01_2025_Jan_Q01

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1. The arithmetic series S is given by

$$S = 2 + 5 + 8 + 11 + \dots + 254$$

Find

(b) the sum of the series.

(2)

Adaptive unit 2 · WMA12-01_2025_Jan_Q01

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1. An arithmetic series starts

$$78 + 75 + 72 + \dots$$

Find

(a) the 18th term in the series,

(2)

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1. An arithmetic series starts

$$78 + 75 + 72 + \dots$$

Find

(b) the sum of the first 100 terms of the series.

(2)

Adaptive unit 2 · WMA12-01_2025_Oct_Q01

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5. A company that owned a silver mine
- extracted 480 tonnes of silver from the mine in year 1
 - extracted 465 tonnes of silver from the mine in year 2
 - extracted 450 tonnes of silver from the mine in year 3

and so on, forming an arithmetic sequence.

(a) Find the mass of silver extracted in year 14

(2)

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5. A company that owned a silver mine
- extracted 480 tonnes of silver from the mine in year 1
 - extracted 465 tonnes of silver from the mine in year 2
 - extracted 450 tonnes of silver from the mine in year 3
- and so on, forming an arithmetic sequence.

After a total of 7770 tonnes of silver was extracted, the company stopped mining.

Given that this occurred at the end of year N ,

(b) show that

$$N^2 - 65N + 1036 = 0 \tag{3}$$

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5. A company that owned a silver mine
- extracted 480 tonnes of silver from the mine in year 1
 - extracted 465 tonnes of silver from the mine in year 2
 - extracted 450 tonnes of silver from the mine in year 3
- and so on, forming an arithmetic sequence.

(c) Hence, state the value of N .

(1)

Adaptive unit 3 · WMA12-01_2021_Oct_Q05

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